

SPH4U Unit 3 Gravitational, Electric, and Magnetic Fields

Chapter 6 Gravitational Fields

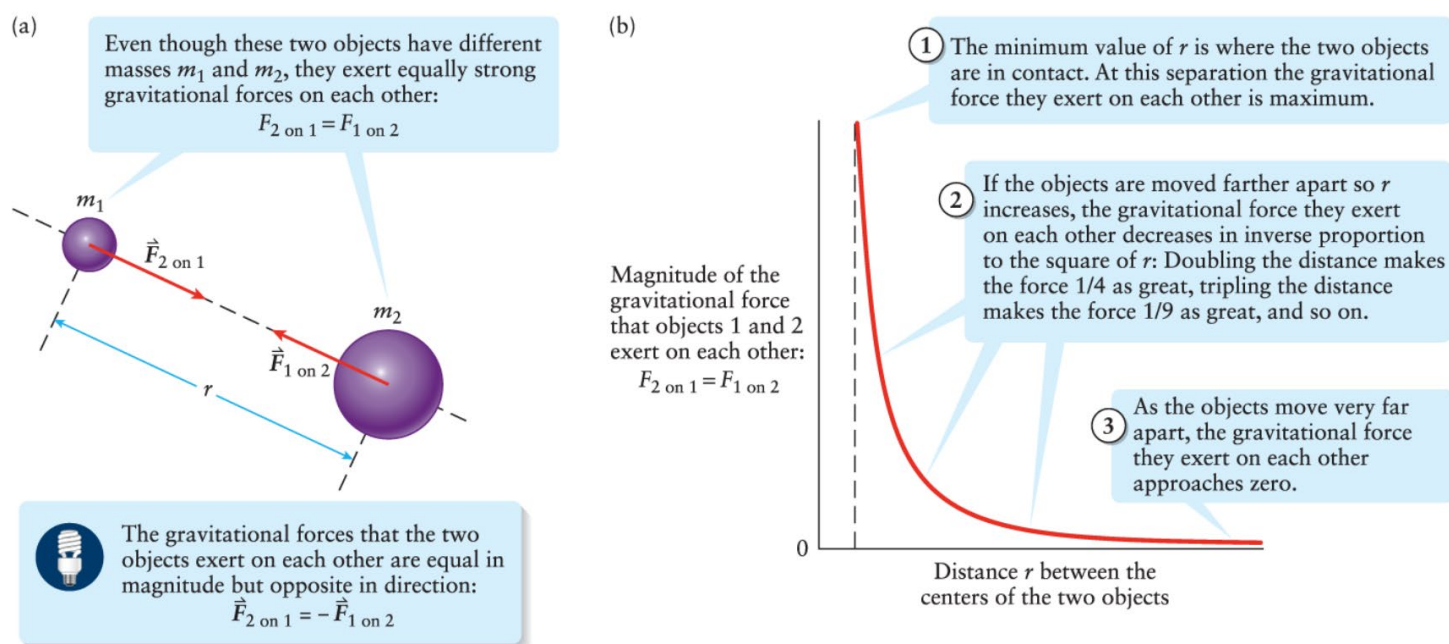
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Lesson 6.1 Newtonian Gravitation

Lesson Goal: The ideas of universal gravitation and circular motion will help us understand the motions of satellites around Earth and of planets around the Sun.

Success Criteria:

- ✓ Describe why an object moving in a circle is always accelerating even when its speed is not changing.
- ✓ Apply Newton's laws to objects in uniform circular motion.
- ✓ Recognize what it means to say that gravitation is universal, and articulate when the gravitational force is the dominant force between objects or systems.
- ✓ Apply Newton's law of universal gravitation to describe or calculate the gravitational force any two objects exert on each other, and use that force in contexts other than orbital motion.
- ✓ Apply the law of universal gravitation to analyze circular orbits of satellites and planets.
- ✓ Relate the gravitational field at a point in space to the gravitational force exerted on an object at that point in space.
- ✓ Explain the origin of apparent weightlessness.
- ✓ Understand properties of the gravitational field
- ✓ Know how to calculate the gravitational field strength



Universal Law of Gravitation

Watch Out! The distance r in the equation is the **center-to-center distance**. Always remember that r in the equation is the distance between the centers of the two objects. If you're calculating the gravitational force that Earth exerts on a person standing on Earth's surface, the distance r equals the radius of Earth—that is, the distance from the center of Earth to that person. For nearly symmetrical objects like the Earth or the Moon, the geometrical center is the same as the center of mass, so we'll often use the terms center and center of mass interchangeably when discussing gravitational forces.

Common Errors & Misconceptions

The fact that two objects will gravitationally attract each other with forces of equal magnitude is very counterintuitive, as students typically assume that the force of attraction exerted by the larger object on the smaller object “should” be larger (for example, students mistakenly believe that Earth exerts a larger force on the Moon than the force that the Moon exerts on Earth). Keep in mind Newton's third law. In particular, it is important to point out that while the forces are indeed equal and opposite, the outcome (that is, acceleration) of these forces need not be the same. Consider a ball released from rest that accelerates due to the force of gravity exerted on it by Earth. The gravitational force of attraction exerted on the ball by Earth is equal to the force exerted on Earth by the ball, but the acceleration of the ball is much greater because its mass is much smaller.

Common Errors & Misconceptions

Do not confuse the acceleration due to gravity, g , with the universal gravitational constant, G . Remember that g is not truly a constant, but rather a variable that depends on location: g is approximately 9.8m/s^2 everywhere near Earth's surface, but the exact value does change slightly from one location to another and is dramatically different on other planets and locations. Near the surface of the Moon, g is approximately 1.6m/s^2 . The gravitational constant G is a universal constant: Its value is the same regardless of the context or location.

Common Errors & Misconceptions

Students tend to think of gravity as a very localized phenomenon, limited to the downward (that is, toward the ground) force exerted on objects that are very near to Earth. Keep in mind that the gravitational force is a universal force between all objects. Why is it difficult to observe the attraction between human-scale objects (that is, due to their relatively low masses)?

Watch Out! What Causes Weightlessness and what doesn't

It's a common misconception that an object in orbit is “beyond the pull of Earth's gravity.” It's true that an astronaut on board the International Space Station floats around the cabin as though she were weightless and unaffected by gravity. But if gravity weren't exerted by Earth on an orbiting satellite or its occupants, the satellite would have zero acceleration and hence couldn't stay in orbit: It would maintain a constant velocity and simply fly off into space in a linear path.

In fact, a satellite such as the ISS remains in orbit precisely because it *does* feel the pull of gravity, which provides the acceleration necessary to make the satellite follow a circular trajectory around Earth. An astronaut on board the ISS seems to be weightless because she is really an independent satellite of Earth following the same orbit as the ISS does. Because the astronaut and the ISS are both following the same trajectory, there is nothing pushing the astronaut toward any of the station's walls. As a result, even though gravity is exerted on the astronaut, she feels weightless.

Since weightlessness means you are accelerating at a rate equal to the acceleration due to gravity, it means you are in free fall. An object in free fall—under only the influence of gravity—feels weightless.

Example 1: Calculating the Force of Gravity between Two Ordinary Masses

The centres of two uniformly dense spheres are separated by 50.0 cm. Each sphere has a mass of 2.00 kg.

- (a) Calculate the magnitude of the gravitational force of attraction between the two spheres.
- (b) How much of an effect will this force have on the two spheres?

Example 2: Calculating the Force of Gravity and Solving for Mass

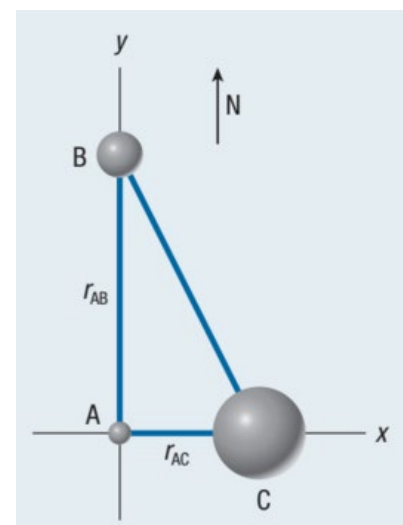
Eris, a dwarf planet, is the ninth most massive body orbiting the Sun. It is more massive than Pluto and three times farther away from the Sun. Eris is estimated to have a radius of approximately 1200 km. Acceleration due to gravity on Eris differs from the value of g on Earth. In this three-part problem, you will explore the force of gravity on the surface of Eris.

- (a) Suppose that an astronaut stands on Eris and drops a rock from a height of 0.30 m. The rock takes 0.87 s to reach the surface. Calculate the value of g on Eris.
- (b) Calculate the mass of Eris.
- (c) Suppose that an astronaut stands on Eris and drops a rock from a height of 2.50 m. Calculate how long it would take the rock to reach the surface.

Example 3: Calculating the Force of Gravity in a Three-Body System

The figure on the right shows three large, spherical asteroids in space, which are arranged at the corners of a right triangle ABC. Asteroid A has a mass of 1.0×10^{20} kg. Asteroid B has a mass of 2.0×10^{20} kg and is 50 million kilometres (5.0×10^{10} m) from asteroid A. Asteroid C has a mass of 4.0×10^{20} kg and is 25 million kilometres (2.5×10^{10} m) away from asteroid A along the other side of the triangle.

- (a) Determine the net force on asteroid A from asteroids B and C.
- (b) Determine the force on asteroid B from asteroid C.



Example 4: Determining the Gravitational Field Strength

(a) Calculate the magnitude of the gravitational field strength on the surface of Saturn, assuming that it is perfectly spherical with a radius of 6.03×10^7 m. The mass of Saturn is 5.69×10^{26} kg.

(b) Determine the ratio of Saturn's gravitational field strength to Earth's gravitational field strength (9.8 N/kg).

Table 2 Surface Gravitational Field Strength of the Planets in the Solar System

Planet	Value of g_{planet} relative to Earth	Value of g (N/kg)
Mercury	0.38	3.7
Venus	0.90	8.8
Earth	1.00	9.8
Mars	0.38	3.7
Jupiter	2.53	24.8
Saturn	1.06	10.4
Uranus	0.90	8.8
Neptune	1.14	11.2

Problem 1 (BFW pg 263) At what height above Earth's surface is the value of g equal to one-half the value at the surface?

Video: [Gravity pulls two metal balls together](#)

Homework

1. TIPERs WWT92: TWO ASTEROIDS—GRAVITATIONAL FORCE ON EACH

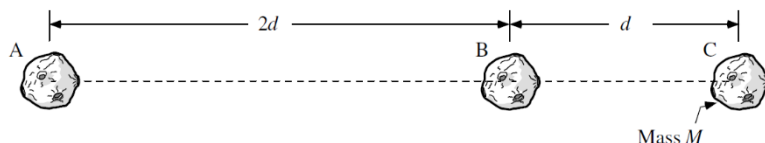
Two asteroids with masses of m and $3m$ exert gravitational forces on each other. A student contends that the forces will be in different directions and of different magnitudes as shown below.



What's wrong, if anything, with this student's contention? If something is wrong, identify it and explain how to correct it. If this student's contention is correct, explain why.

2. TIPERs SCT93: THREE ASTEROIDS IN A LINE—CALCULATION OF MASS

At the instant shown, three asteroids are in a line, and the distance between A and B is twice the distance between B and C. Asteroid C has mass M . There is no net force on asteroid B due to the other asteroids.



Three students are discussing how they might find the mass of asteroid A:

Ari: "We don't really have enough information to find the mass of A. Since there's no net force on B, the force from A has to cancel the force from C. To find the force on B from C, we'd use Newton's law of universal gravitation. But since the force is proportional to the product of the masses, we'd need to know both masses."

Bira: "I don't think we really need the mass of B. Asteroid A is twice as far away as C, so if it also has a mass M it will exert half as much force as C does. Since it has to exert the same force for the net force on B to be zero, it has to have twice the mass."

Cole: "It's true that A pulls on B to the left, and C pulls on B to the right. But you can't just use Newton's law of universal gravitation, because that only allows you to calculate the force between two masses. Here there are three masses, and asteroid A is exerting some of its force on B and some on C."

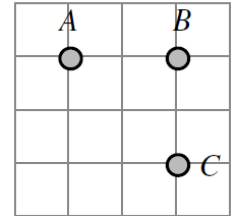
With which, if any, of these students do you agree? *Ari* _____ *Bira* _____ *Cole* _____ None of them _____

Explain your reasoning.

3. TIPERs QRT94: THREE OBJECTS EXERTING GRAVITATIONAL FORCES—NET FORCE

Three objects, each with a mass of M , exert gravitational forces on each other. Which of the arrows below shows the direction of the net force on mass B ?

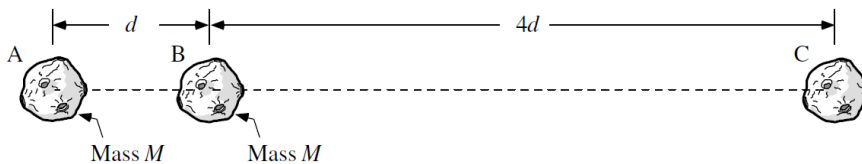
- (i)  (ii)  (iii)  (iv)  (v) None of these



Explain your reasoning.

4. TIPERs WWT95: THREE ASTEROIDS IN A LINE—MASS OF ASTEROID

At the instant shown, three asteroids are in a line, and the distance between B and C is 4 times as large as the distance between A and B . Asteroids A and B have the same mass. There is no net force on asteroid B due to the other two asteroids.



A student makes the following comment about the mass of asteroid C :

“Since C is four times as far from B as A is, it is only going to have one-quarter the effect on B . To get the forces on B to balance, you’d need the mass of C to be four times as large.”

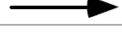
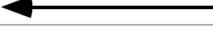
What, if anything, is wrong with the student’s statement? If nothing is wrong, state that explicitly, and explain why it is correct. If the statement is incorrect, state what is wrong and how you would correct it.

5. TIPERs QRT96: TWO OBJECTS—GRAVITATIONAL FORCE ON EACH

Object A has twice the mass of Object B.

Identify the pair of force vectors (the arrows) that correctly compare the gravitational force exerted on A by B with the gravitational force exerted on B by A.

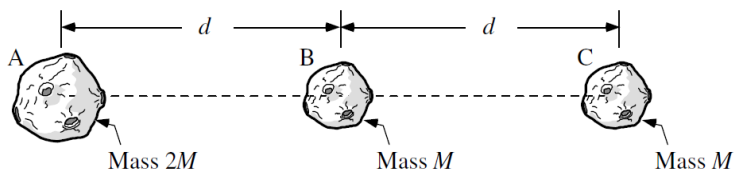


	Gravitational force on A by B	Gravitational force on B by A
(a)		
(b)		
(c)		
(d)		
(e)		
(f)		

Explain your reasoning.

6. TIPERs RT97: THREE ASTEROIDS IN A LINE—NET FORCE

At the instant shown, three asteroids are in a line, and the distance between A and B is the same as the distance between B and C. Asteroids B and C have the same mass, while asteroid A has twice the mass.



Rank the magnitude of the net force on each asteroid due to the other two asteroids.

			OR			
1	2	3		All the same	All zero	Cannot determine
Greatest		Least				

Explain your reasoning.

Check Your Understanding - Lesson 6.1

1. Do objects of different mass fall at different rates? (
2. Perform the following thought experiment based on gravitational forces. Imagine an insulated, hollow tunnel cut straight through Earth, passing through the centre. What would happen to a ball tossed into the tunnel?

6.2 Orbits

Lesson Goal: Understanding the universal law of Gravitation and properties of the gravitational field.

Success Criteria:

- ✓ Apply the law of universal gravitation to analyze circular orbits of satellites and planets.
- ✓ Explain the origin of apparent weightlessness
- ✓ Know to calculate the speed of a satellite in a uniform circular motion
- ✓ Understand that a satellite in a circular orbit with a given orbital radius has a constant speed

Satellites in Circular Orbits

Orbital radius _____

Video: [ISS – Chris Hadfield](#)

Example 1: Calculating the Speed and Orbital Period of a Satellite

The International Space Station (ISS) orbits Earth at an altitude of about 350 km above Earth's surface.

- (a) Determine the speed needed by the ISS to maintain its orbit.
- (b) Determine the orbital period of the ISS in minutes.

Example 2: Calculating the Speeds of Planets around the Sun

Determine the speeds of Venus and Earth as they orbit the Sun. The Sun's mass is 1.99×10^{30} kg. Venus has an orbital radius of 1.08×10^{11} m, and Earth has an orbital radius of 1.49×10^{11} m.

Problem #1 (AP chapter 6) Students are to carry out a series of experiments with blocks in a lab. For the first experiment, some students decide to use a launcher to give both blocks the same initial velocity, v_0 , and launch them across a table. The coefficient of kinetic friction between the table and the blocks is μ_k . The blocks are made of the same material, but the mass of block A, m_A , is twice the mass of block B, m_B .

- a) i. Student 1 says block A will slide farther along the table before coming to rest because it has more inertia. What is correct or incorrect about this statement? Explain briefly. (Do not use an equation; state your reasoning conceptually.)
- ii. Student 2 says block B will slide farther because the floor exerts a smaller normal force on it. What is correct or incorrect about this statement? Explain briefly. (Do not use an equation; state your reasoning conceptually.)
- b) Derive an expression for the distance travelled by block A after it leaves the launcher in terms of given quantities and physical constants.
- c) Use the expression you derived in part (b) to evaluate the correctness of the student statements above. Explain briefly.

Problem #3 (BFW) - Satellite Motion Comparison Task

Two satellites are in stable, circular orbit around a distant planet. Satellite A is orbiting at a radius R_A and has mass M_A . Satellite B is orbiting at radius $R_B = 2R_A$ and has mass $M_B = M_A/2$.

- a) Which of the two satellites has a greater speed? Justify your claim.
- b) Which of the two satellites has a greater acceleration? Justify your claim.
- c) Which of the two satellites has a greater orbital period? Justify your claim.

Check Your Understanding – Lesson 6.2

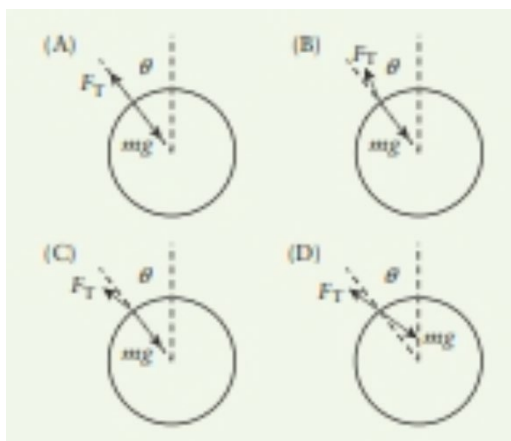
1. Is gravity in space zero?

Homework

MC #1 (BFW – AP Practice Problem ch 6) Consider a nonrotating planet of radius R that has no atmosphere. On Earth, the atmosphere helps keep everything on the surface in place as the planet rotates. Because it has no atmosphere, on this other planet just the gravitational force exerted by the planet keeps things on the surface. Take the acceleration of gravity on the surface of that planet as g_p . Now consider a mass m located on the surface at the planet's equator. How fast would the planet have to rotate at its surface for the object's apparent weight to be reduced to three-quarters of that for the nonrotating planet?

$$\begin{array}{ll} \text{(A)} \ v = \frac{\sqrt{g_p R}}{2} & \text{(B)} \ v = \frac{\sqrt{3g_p R}}{2} \\ \text{(C)} \ v = 2\sqrt{g_p R} & \text{(D)} \ v = \frac{3\sqrt{g_p R}}{4} \end{array}$$

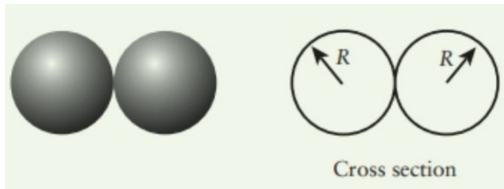
MC #2 (BFW #4 – AP Practice Problems ch 6) Consider an object of mass m located at an angle θ from the axis of rotation of a planet. It is hanging from a string. Of the following, which diagram best shows the relationship between the force of gravity, mg , and the tension in the string, F_T , on that rotating planet?



MC #3 (BFW #7 – AP Practice Problems ch 6) The net gravitational force exerted on an inter-stellar traveler by two stars is zero. What conclusion can be drawn regarding the location of the traveller relative to the two stars? Choose the best answer.

- A) The two stars are the same distance from the traveler.
- B) The ratios of the mass of each star to the distance from the star to the traveler are the same.
- C) The ratios of the mass of each star to the distance squared from the star to the traveler are the same.
- D) The ratios of the mass of each star to the distance squared from the star to the traveller are the same, and the traveller is located on a line connecting the centers of the two stars.

Questions 4 and 5 refer to the following material.



MC #4 (BFW #8 – AP Practice Problems ch 6) Two identical solid steel spheres of radius R are placed in contact, as shown in the figure above. The magnitude of the gravitational force that they exert on each other is F_1 . Two different solid spheres, each of radius $3R$, but made of this same steel, are placed in contact with each other. In terms of the original force F_1 , what is the magnitude of the gravitational force that these new spheres exert on each other?

- A) F_1
- B) $3F_1$
- C) $9F_1$
- D) $8F_1$

MC #5 (BFW #9) Two solid identical steel spheres of radius R are placed in contact, as shown in the figure above. The magnitude of the gravitational force that they exert on each other is F_1 . Two solid identical brass spheres of the same radius R (brass is denser than steel) are placed in contact. The magnitude of the gravitational force that they exert on each other is F_2 . Two solid identical aluminum spheres of the same radius R (aluminum is less dense than steel) are placed in contact. The magnitude of the gravitational force that they exert on each other is F_3 . How do the magnitudes of the gravitational forces compare?

- A) $F_1 = F_2 = F_3$
- B) $F_1 > F_2 > F_3$
- C) $F_2 > F_1 > F_3$
- D) $F_3 > F_1 > F_2$

MC #6 (6-5 pg 271 #5) to Newton's universal law of gravitation, if the masses of both the object exerting the force and the object on which the force is exerted are doubled, the force is

- A) four times as large as the original value.
- B) twice as large as the original value.
- C) one-half of the original value.
- D) one-fourth of the original value.

MC #7 (6-5 pg. 271 #7) Two planets, A and B, with the same mass are in circular orbits around the same star. The orbital radius of A is four times larger than the orbital radius of B. The period of planet B is

- A) eight times the period of planet A.
- B) four times the period of planet A.
- C) one-fourth of the period of planet A.
- D) one-eighth of the period of planet A.

MC# 8 (6-5 pg. 272 #8) According to Newton's universal law of gravitation, if the distance r between two objects is doubled, the force exerted by one of these objects on the other is

- A) four times as large as the original value.
- B) twice as large as the original value.
- C) one-half of the original value.
- D) one-fourth of the original value.

MC #9 (BFW) Satellite A is in a stable circular orbit above Earth's surface at a radius R_A and satellite B is in orbit at a radius $R_B=2R_A$. Which of the following is the correct relationship between their orbital speeds, v_A and v_B , respectively?

- A) $v_A/v_B = 1/2$
- B) $v_A/v_B = 2$
- C) $v_A/v_B = \sqrt{2}$
- D) $v_A/v_B = 1/\sqrt{2}$

Additional Problems

Problem 1 (pg. 267 #4) No net force is exerted on a space probe located at some distance $r_{\text{Earth-probe}}$ along a line joining the center of Earth and the center of the Moon. Representing the center-center Earth-Moon distance as $r_{\text{Earth-Moon}}$, express the condition of no net force as a quadratic equation whose solution is the distance $r_{\text{Earth-probe}}$.

Problem 2 (pg. 267 #7) We observe that the Moon revolves around Earth once every 27.3 days. The average distance from the center of Earth to the center of the Moon is 3.84×10^8 m. Determine the acceleration of the Moon due to its motion around Earth assuming that Earth is stationary and the Moon is in uniform circular motion.

Problem 3 (pg. 267 #8) Commercial ultracentrifuges can rotate at rates of 100,000 rpm (revolutions per minute). A particular centrifuge produces a centripetal acceleration of 800,000 g. Calculate the distance from the rotation axis to the sample chamber in this device.

Problem 4 (pg. 271 #3): A satellite is to be raised from one circular orbit to one farther from the Earth's surface. What will happen to its period?

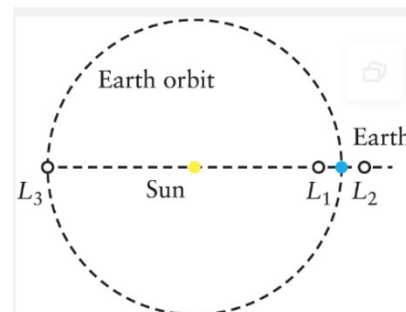
Problem 5 (pg. 271 #4) A geostationary satellite orbits in the equatorial plane of Earth and has a unique orbital radius and speed.

- Describe in complete sentences why the radius and speed are unique.
- Imagine that a geosynchronous satellite orbits in a plane intersecting Earth's surface at a latitude θ (the angle above Earth's equator). Construct a diagram showing the plane and radius of the satellite from Earth's center, r , and the radius of its orbital path if it were to travel in a circle.
- It turns out that a geosynchronous satellite's orbit must lie in a plane through Earth's center. Write down the equation for gravitational force and that for mass times centripetal acceleration, and identify what the r 's are and why the relationships previously used don't work if this isn't the case.

Problem 6 (pg. 272 #9) Consider the following figure. The net gravitational force on a satellite placed at point L_1 or L_2 causes the satellite to orbit the Sun with the same period as Earth, T_{Earth} . You may assume the satellites have just two forces exerted on them: one by Earth and one by the Sun.

- Using free-body diagrams and Newton's second law, show that the centripetal acceleration, a_{L1} of an object at the point L_1 and the centripetal acceleration, a_{L2} of an object at the point L_2 can be expressed as

$$-\frac{GM_{\text{Sun}}}{r_{\text{Sun-L1}}^2} + \frac{GM_{\text{Earth}}}{r_{\text{Earth-L1}}^2} = a_{L1} \quad \text{and} \quad -\frac{GM_{\text{Sun}}}{r_{\text{Sun-L2}}^2} - \frac{GM_{\text{Earth}}}{r_{\text{Earth-L2}}^2} = a_{L2}$$



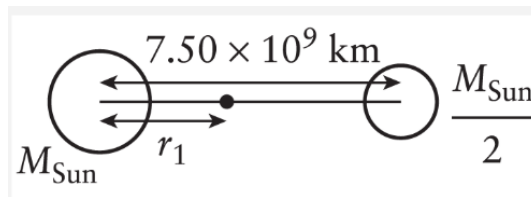
b) Explain why the orbital period of L_1 and Earth will be equal if

$$-\frac{M_{\text{Sun}}}{r_{L_1-\text{Sun}}^3} + \frac{M_{\text{Earth}}}{r_{\text{Earth}-L_1}^2 r_{\text{Sun}-L_1}} = -\frac{M_{\text{Sun}}}{r_{\text{Earth}-\text{Sun}}^3}$$

This expression can be used to calculate the distance from Earth to L_1 . The distance is approximately 1.5×10^9 m, to be compared with $r_{\text{Earth}-\text{Sun}} = 1.5 \times 10^{11}$ m.

c) Euler described $r_{L_1-\text{Earth}}$ and $r_{L_2-\text{Earth}}$ in 1765. He also described a third point, L_3 , with the same orbital period as Earth, T_{Earth} . Explain why the periods of L_3 and Earth must be the same.

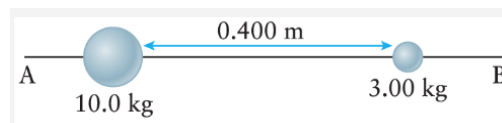
Problem #7 (Ch 6 Review pg. 278 #12) Two stars are separated by a center-to-center distance of 7.50×10^9 km. The first star has a mass equal to the mass of our Sun, $M_1 = M_{\text{Sun}}$. The second star has a mass that is one-half of the Sun's mass, $M_2 = M_{\text{Sun}}/2$. Let r_1 be the distance from the center of the first star to a point along the line between the stars, as shown.



a) Construct a free-body diagram for an object with mass m instantaneously at rest at point r_1 .

b) Construct but do not solve a mathematical equation in terms of M_{Sun} , r_1 , and any necessary constants that you could use to solve for r_1 , a distance such that the object remains at rest at that point.

Problem #8 (Ch 6 Review pg. 279 #28) Locate the point(s) along the line AB where a small, 1.00-kg object could be placed such that the net gravitational force on it due to the two objects shown is exactly zero.



Problem #9 (Ch 6 Review pg. 279 #32) Two satellites are in stable, circular orbits about a planet. One is at a radius R_A and the other is at a radius $R_B = 2R_A$.

a) Calculate the ratio of the speeds of these satellites in their orbit, v_B/v_A .

b) Calculate the ratio of the periods of these satellites, T_B/T_A .

Chapter 6 - Check Your Knowledge – Definitions

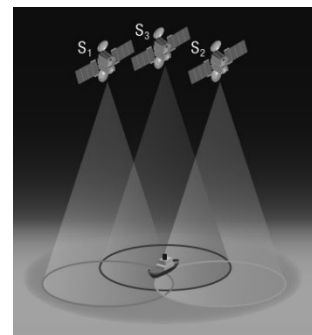
Term	Definition
____ gravitational field	1. a spacecraft in which people live and work
____ satellite	2. an object that has been intentionally placed by humans into orbit around Earth or another body; referred to as “artificial” to distinguish from natural satellites such as the Moon
____ geosynchronous orbit	3. the distance between the centre of a satellite and the centre of its parent body
____ gravitational field strength	4. a mathematical relationship in which one variable is proportional to the inverse of the square of another variable; the law applies to gravitational forces and other phenomena, such as electric field strength and sound intensity
____ orbital radius	5. an object or a body that revolves around another body due to gravitational attraction
____ gravitational constant	6. the orbit around Earth of an object with an orbital speed matching the rate of Earth’s rotation; the period of such an orbit is exactly one Earth day
____ inverse-square law	7. a collection of vectors, one at each point in space, that determines the magnitude and direction of the gravitational force
____ artificial satellite	8. a constant that appears in the universal law of gravitation; the constant is written as G and has a value of $6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$
____ space station	9. the magnitude of the gravitational field vector at a point in space

Chapter 6 –Communication Questions

1. The planet Alpha has a radius r and a mass m . The planet Bravo has twice the mass, but the same surface gravity. What is the radius of planet Bravo? Explain your answer.
2. Why can a gravitational field vector never point away from Earth?
3. How would you convince a relative who had not studied physics that the moon is really a falling body?
4. When a star like the Sun has exhausted most of its fuel, it expands into a red giant, with a radius approximately equal to the orbit of Earth, an increase of about 216 times. By what factor does its surface gravity change, assuming that the mass remains constant?
5. You are located a distance r from a planet with mass m , and feel a gravitational force F_g . Later, you find yourself a distance of $4r$ from a planet with a mass of $8m$. How does the gravitational force from the second planet compare with that from the first?
6. Mikhail says, “since the mass of Earth is much greater than my mass, the force that Earth exerts on me is much greater than the force I exert on Earth.” Is he correct? Explain your answer.
7. Explain why Henry Cavendish used a light beam to measure the gravitational force between two lead spheres rather than a spring balance.
8. When working out physics problems involving the gravitational acceleration at the surface of Earth, the value of g is usually given to 2 or at most 3 significant digits, although modern methods of measurement allow much greater accuracy. Why is this?
9. Under what conditions may you use the equation
$$v = \sqrt{\frac{Gm_E}{r}}$$
 for a satellite?
10. Why are some satellites put into geostationary orbits while others are not?
11. Exoplanets, i.e., planets circling other stars than the sun, are being found on a daily basis. If evidence of intelligent life is found on one of these planets, how could this affect your view of the world? What beliefs might you need to change?

12. An amusing science-fiction short story by Jerome Bixby features satellites that orbit Mars just above the surface. They are fragments of very dense matter similar to that in a white dwarf star. How could such satellites exist in such a low orbit? Is it possible on Mars? Would this be possible on Earth?

13. A ship on the ocean can locate its position using signals from three GPS satellites. Would this work for an airplane in flight? Explain your answer.

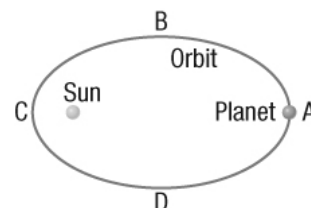


14. The extinction of the dinosaurs is attributed by some scientists to the results of an asteroid colliding with Earth. How does the material in this chapter fit in with this interpretation of extinction in a way that impacts your own future? Use the Internet to determine how likely this is.

15. An astronomer has observed an exoplanet orbiting a nearby star, and determined the period of orbit of the planet, and its orbital radius around the star. Is this enough information to determine the mass of the planet? If not, what other information would the astronomer need? Explain your answer.

16. Xenia says, “The orbital speed of a satellite would remain unchanged if the central body’s mass and radius were increased by the same factor k .” Is she correct? Explain your answer.

17. Planetary orbits are not perfect circles, but rather ellipses. The Sun is not at the centre of the ellipse, but at a special point known as a focus, as shown. Describe the changes in speed of a planet as it makes one complete orbit.



18. The GPS system uses a constellation of 24 satellites. A newer version of GPS promises better coverage, and a strong enough signal to be used inside a building or in a forest without a clear view of the sky. How will this impact your life? Select one or two ways that this new technology will affect you personally.
19. Satellites show that the ice caps in Greenland and the Antarctic are shrinking. These ice caps contain almost $\frac{3}{4}$ of the fresh water on the surface of Earth. How will the melting of the caps affect you personally? What action can you take to help stop this phenomenon?